MILITARY SPECIFICATION

BATTERY, DRY, FOR FIRING DEVICE: M122

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers quality assurance provisions and preparation for delivery for one type of battery for the M122 Firing Device Transmitter.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.2.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For.

MIL-B-55521 - Battery, Dry Packaging and Packing of.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1375

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.5)

U. S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9296723 - Battery, Dry, For Firing Device: M122

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings, specifications, and standards.

3.2 Batteries. The batteries shall comply with all requirements specified on (Dwg.) 9296723, all associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Electrolyte leakage. There shall be no evidence of electrolyte leakage on the external surfaces of the jacket at any time prior to or during the performance of test specified in 4.5.1 or any other tests specified herein. Electrolyte leakage shall be considered to have occurred when a drop of phenolphthalein solution placed on the battery turns pink or red.

3.4 Open circuit voltage. The battery shall have an open circuit voltage of eighteen volts plus two and seven tenths volts (18.0V $\pm$ 2.7V) at a temperature of seventy degrees fahrenheit plus or minus ten degrees fahrenheit ($70^{\circ} + 10^{\circ}F$).
$- 0$ V

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3.5 Closed circuit voltage. The battery shall have a minimum closed circuit voltage of sixteen and four tenths (16.4) volts at a temperature of seventy degrees Fahrenheit plus or minus ten degrees Fahrenheit ($70^{\circ}F \pm 10^{\circ}F$) when measured across a resistive load of twenty-five plus or minus two (25 ± 2) olms.

3.6 Insulating, impregnating, potting and sealing compounds. The insulating, impregnating, potting and sealing compounds shall exclude moisture from insulating material without impairing its electrical characteristics. When tested as specified in 4.5.2, the potting and sealing compounds shall not flow at high temperature ($160^{\circ}F$) nor crack or draw away from the sides of a container at low temperature ($-40^{\circ}F$), or during any temperature change between $+160^{\circ}F$ and $-40^{\circ}F$ sufficiently to impair electrical connections.

3.7 Contact resistance. Contact resistance between each socket insert and the applicable pin of the mating-plug shall not exceed five thousandths (0.005) ohm.

3.8 Waterproofing of jacket (WOJ). The jacket shall not fall apart and the seams shall remain intact when tested as specified in 4.5.4.

3.9 Color of jackets. The color of exposed surfaces of jackets shall match one of the following lusterless greens 34079, 34086, 34087, 34096, 34102, 34127, and 34128 per Federal Standard No. 595.

3.10 Capacity.

3.10.1 Low temperature capacity. The battery shall be stored for a minimum period of sixteen (16) hours at a temperature of zero degrees Fahrenheit minus ten degrees Fahrenheit ($0^{\circ}F \begin{smallmatrix}+0^{\circ}F\\-10^{\circ}F\end{smallmatrix}$) prior to discharge. After storage the battery shall be discharged across a resistive load of two hundred plus or minus ten (200 ± 10) ohms at a temperature of zero degrees Fahrenheit minus ten degrees Fahrenheit ($0^{\circ}F \begin{smallmatrix}+0^{\circ}F\\-10^{\circ}F\end{smallmatrix}$). The battery shall remain above ten and eight tenths (10.8) volts during discharge for a minimum of eight (8.0) hours.

3.11.2 Ambient temperature capacity. The battery shall be stored for a minimum period of forty-eight (48) hours at a temperature of seventy degrees Fahrenheit plus or minus ten degrees Fahrenheit ($70^{\circ}F \pm 10^{\circ}F$) prior to discharge.

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After storage the battery shall be discharged across a resistive load of two hundred plus or minus ten (200 ± 10) ohms at a temperature of seventy degrees Fahrenheit plus or minus ten degrees fahrenheit (70°F ± 10°F).

The battery shall remain above ten and eight tenths (10.8) volts during discharge for a minimum of forty (40) hours.

3.10.3 High temperature capacity. The battery shall be stored for a period of seven plus one (7 + 1) days at a temperature of one hundred and sixty degrees Fahrenheit plus ten degrees fahrenheit (+160°F + 10°F
- 0°F).
The battery shall then be stored for a minimum period of forty-eight (48) hours at a temperature of seventy degrees Fahrenheit (70°F ± 10°F).

After storage the battery shall be discharged across a resistive load of two hundred plus or minus ten (200 ±10) ohms at a temperature seventy degrees Fahrenheit plus or minus ten degrees Fahrenheit (70°F ± 10°F).

The battery shall remain above ten and eight tenths (10.8) volts during discharge for a minimum of thirty-five (35) hours.

3.11 Vibration. The battery after being tested for vibration in accordance with 4.5.9 shall empty with the requirements of 3.5 and 3.7.

3.12 Mechanical shock. The battery after being tested for mechanical shock in accordance with 4.5.10 shall comply with the requirements of 3.5 and 3.7.

3.13 Insulation resistance. The insulation resistance between all terminals and the container of the battery, shall be not less than 5 megohms when tested as specified in 4.5.8.

3.14 Age of cells. The minimum age of cells, from the time of their fabrication to the time of their presentation for acceptance inspection as batteries, shall be 5 days. Batteries shall be submitted for acceptance inspection not more than 30 days prior to the shipping date. The cells shall be checked as specified in 4.5.11.

3.15 Labeling and marking. All labeling and marking shall be clear and legible throughout all of the tests specified herein.

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3.16 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.17 Workmanship. Batteries shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect their life, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection (See 4.3)

b. Quality conformance inspection (See 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of one hundred twenty (120) batteries. (Dwg. 9296723)

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. FIRST ARTICLE INSPECTION.

CLASSIFICATION OF DEFECTS & TESTS MIL-B-63116B(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Battery, Dry, For Firing Device: M122				9296723
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Examination for Defects	25		3.2	4.4.2.1
	Tests				
	Electrolyte leakage	10		3.3	4.5.1
	Insulating, impregnating, potting and sealing compounds	10		3.6	4.5.2
	Waterproofing of jacket	10		3.8	4.5.4
	Vibration	10		3.11	4.5.9
	Mechanical Shock	10		3.12	4.5.10
	Insulation resistance	25		3.13	4.5.8
	Contact resistance	25		3.7	4.5.3
	Open circuit voltage	25		3.4	4.5.5
	Closed circuit voltage	25		3.5	4.5.6
	Low temperature capacity	25		3.10.1	3.7
	Ambient temperature capacity	25		3.10.2	3.7
	High temperature capacity	25		3.10.3	3.7

NOTES:
1.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plan and procedures for major and minor defects shall be in accordance with MIL-STD-105.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-63116B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Battery, Dry, For Firing Device: M122		9296723
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length		0.40%	3.2	Gage
102	Width		0.40%	3.3	Gage
103	Height		0.40%	3.2	Gage
104	Electrolyte leakage	10/0/1		3.3	4.5.1
105	Insulating, impregnating, potting	10/0/1		3.6	4.5.2
106	Waterproofing of jacket	10/0/1		3.8	4.5.4
107	Weight	10/0/1		3.2	Balance
108	Color of jacket	10/0/1		3.9	Visual
109	Labeling	10/0/1		3.15	Visual
111	Vibration	10/0/1		3.11	4.5.9
112	Mechanical shock	10/0/1		3.12	4.5.10
113	Age of cells		100%	3.14	4.5.11
114	Insulation resistance		100%	3.13	4.5.8
115	Contact resistance		100%	3.7	4.5.3
116	Open circuit resistance		100%	3.4	4.5.5
117	Closed circuit voltage		100%	3.5	4.5.6
118	Low temperature capacity	10/0/1		3.10.1	3.7
119	Ambient temperature capacity	10/0/1		3.10.2	3.7
120	High temperature capacity	10/0/1		3.10.3	3.7
Minor					
201	Evidence of poor workmanship		1.0%	3.17	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6. herein.

4.5 Test methods and procedures.

4.5.1 Electrolyte leakage. The battery shall be tested for the specified requirement by having a drop of phenolphthalein solution placed on the sample. If the solution fully or partially turns pink or red, the battery shall be classed defective. Sample batteries shall be stored for 15 days. Fifty percent of the batteries shall be stored in an inverted position. During the 15-day storage period, each battery shall be examined daily for evidence of electrolyte leakage on the external surfaces of the jacket. Any battery failing to meet the specified requirements shall be classed defective. (Destructive test).

4.5.2 Insulating, impregnating, potting and sealing compound. Potting and sealing compounds, when used, shall be placed in a container, approximately 3 inches wide by 6 inches long by 3/4 inch high, to within 1/4 inch of the top. The temperature of the compound within the container shall be raised to $160^{\circ} \pm 5^{\circ}F$ ($71.1^{\circ} \pm 2.8^{\circ}C$) and the container shall be held in an inverted position for 24 hours. Then the temperature of the compound shall be lowered to $-40^{\circ} \pm 5^{\circ}F$ ($-40^{\circ} \pm 2.8^{\circ}C$). The flow, cracking or shrinking of the compound from the sides of the container shall be noted and the batteries shall be examined to determine that there is no impairment of electrical contact. If flow is noted at $160^{\circ} \pm 5^{\circ}F$ ($77.1^{\circ} + 2.8^{\circ}C$) five batteries of type involved shall be exposed to $160^{\circ} \pm 5^{\circ}F$ ($77.1^{\circ} \pm 2.8^{\circ}C$) storage for 24 hours with terminals resting in lowest possible position. At the end of this exposure period batteries shall be examined to determine that there is no impairment of electrical contact. (See 3.5) Any battery failing to meet the specified requirement shall be classed defective. (Non-destructive test)

4.5.3 Contact resistance. Contact resistance between each socket insert and the applicable pin of the mating-plug shall be measured after insertions while a current of 5 amperes is flowing

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through the point of contact. Any battery failing to meet the specified requirements shall be classed defective. (Non-destructive test)

4.5.4 Waterproofing of jackets (WOJ). All multicell batteries and paper jacketed single cell batteries shall be immersed to within 1/4 inch of the top of the jacket in water maintained at a temperature of 160° ± 5°F (71.1° ± 2.8°C) for a period of 48 hours. In case of multicell metallic jacketed batteries the metal containers without covers may be subjected to this test in place of assembled batteries. In the case of nonmetallic jacketed batteries the jackets may be applied to dummy batteries in place of actual batteries. Condition of the jacket and its seams shall be observed. Any battery failing to met the specified requirements shall be classed defective. (Destructive test)

4.5.5 Open-circuit voltage. A direct current voltmeter of proper range and sensitivity shall be used to measure the open-circuit voltage. Any battery failing to meet the specified requirements shall be classed defective. (Non-destructive test)

4.5.6 Closed-circuit voltage. A direct current voltmeter of proper range and sensitivity shall be used to measure the closed-circuit voltage utilizing resistance specified. Any battery failing to meet the specified requirements shall be classed defective. (Non-destructive test)

4.5.7 Capacity. The battery, after storage at the specified temperature conditions shall be discharged at the specified temperature. During discharge, the battery shall be continuously monitored for the unit under test's output voltage. The test equipment shall be capable of recording the discharge characteristics and temperature at lease once every fifteen (15) minutes of test. Any device failing to comply with all of the specified requirements shall be classed defective. (Destructive test)

4.5.8 Insulation resistance. Insulation-resistance test shall be performed. Batteries shall be stored for a period of 48 hours at 70° ± 5°F (21.1 ± 2.8°C) and a relative humidity of 50 ± 15 percent. After storage and while at these conditions, the insulation resistance shall be measured by applying a direct current potential of 500 ± 20 volts between any two terminals not electrically connected and between all ungrounded terminals and the container of the battery. The insulation resistance of batteries having nonmetallic container shall be measured by the use of a 1 inch-square copper plate making physical contact with the

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container. The plate shall be placed with the broad surface against any area of any surface of the jacket other than that on which the battery terminals are located. Any battery failing to meet the specified requirements shall be classed defective. (Non-destructive test)

4.5.9 Vibration. Vibraton test shall be performed. Each battery shall be rigidly clamped to the platform of a vibration machine in a manner approximating as closely as practicable the manner in which the batteries are clamped when in use. A simple harmonic motion shall be applied having an amplitude of 0.03 inch (0.06 - inch total maximum excursion). The frequency shall be varied at the rate of 1 cycle per second per minute between the limits of 10 and 55 cycles per second. The entire range of frequencies and return shall be traversed in 95 ± 5 minutes for each mounting position, (direction of vibration) of the battery. The batteries shall be vibrated in three equal periods in mutually perpendicular directions, one of which shall be perpendicular to the terminal face of the battery. Open-circuit voltage 3.4.1 shall be observed for 30 seconds during the last quarter of each of the three vibration periods. After vibration the batteries shall meet the specified requirements of 3.4 and 3.7. Any battery failing to meet the specified requirements shall be classed defective. (non-destructive test)

4.5.10 Mechanical shock. Mechanical shock test shall be performed except as otherwise specified. Each battery shall be secured to the testing machine by means of a rigid mount which will support all mounting surfaces of the battery. Each battery shall be subjected to a total of three shocks of equal magnitude. The shocks shall be applied in each of three mutually perpendicular directions. Each shock shall be supplied in a direction normal to a face of the battery. The faces of the battery are identified by their position in relation to the front face (the face which bears the label). For each shock, the battery shall be accelerated in such a manner that during the first 3 milliseconds the minimum average acceleration is 75 gravity units (G). The peak acceleration shall be between 125 to 175 G. After mechanical shock test the batteries shall meet the specified requirements.of 3.4 and 3.7. Any battery failing to meet the specified requirements shall be classed defective. (Destructive test)

4.5.11 Age of cells. The age of the battery cells shall be recorded and checked at the intervals specified in 3.13. Any battery failing to meet the specified requirements shall be classed defective (non-destructive test).

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5. PACKAGING

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with Military Specification MIL-B-55521. Preservation and packaging shall be level A or C with each unit packaging containing one battery, and packing shall be level A, B, or C, as specified in contract or order.

6. NOTES

6.1 Intended use. The components and assemblies covered by this specification are intended for use as the battery for the M122 firing device transmitter.

6.2 Ordering data. See Section 6.1 and 6.2 of MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. (See MIL-A-48078). Submit equipment designs as required to: Commander, ARRADCOM, Dover, NJ 07801, ATTN: DRDAR-QAT-I.

6.4 Distributing of ammunition data cards, inspection results and test data. Distribution of ammunition data cards, inspection and test results shall be submitted to: Commander, ARRADCOM, Dover, NJ 07801, ATTN: DRDAR-QAT-M.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading, "U.S. Army Armament Research and Development Command (ARRADCOM)" may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activites in now under the cognizance of ARRADCOM.

6.6 Age of cells. The maximum age of cells, from the time of their fabrication to the time of their shipping date, shall be 90 days.

6.7 Storage. Sample batteries (packaged per contract) shall be stored at applicable storage conditions for the specified period. (See 3.5) The storage period shall be considered to have started from the date the batteries are actually placed in storage. Any battery failing to meet the specified requirements shall be classed defective. (Non-destructive test)

6.8 Changes from previous issue. Asterisks are not used due to the extensiveness of the changes.

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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